Dylan Becchetti

Engineering Systems | Python, Validation & Workflow Automation

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Portfolio: formsby-systems.github.io/engineering-portfolio

PROFILE

Mechanical engineering graduate of Lehigh with three manufacturing internships and independent work in Python validation and workflow software. Available January 1, 2027 for full-time U.S. remote work.

EXPERIENCE

Cardinal Glass Industries | St. Louis Park, MN

Continuous Improvement Engineering Intern, summers 2024–2025 | Engineering Intern, summer 2023

- Built an Excel macro tool that consolidated separate R&D production files into a daily part-status tracker for shift debriefs. Defined the view with engineers and operators.
- Wrote illustrated instructions for customer sample assemblies and worked on spare-parts storage and inventory. Investigated cleanroom airflow with a micromanometer and hood.
- Inspected pilot-line equipment, analyzed data in Minitab, drafted AutoCAD equipment layouts and contributed to design failure mode and effects analysis discussions.

Operalith | Founder & Product Lead, independent project

- Founded and built a Next.js workspace with five tools for scheduling, projects, inventory, equipment loans and enquiries. Linked the overview to exact records across the tools.
- Designed PostgreSQL saves to return the original result for retries and reject stale edits. A hosted beta test confirmed an edit survived reload and undo preserved all 11 earlier history entries.

SELECTED ENGINEERING PROJECTS

Midas Futures | Creator & Technical Lead

- Conceived and directed a year of daily development, evolving an equities scheduler into Python futures research and paper execution infrastructure with offline replay, approval gates and order and fill reconciliation.
- Unified replay and forward rules in a completed-bar state machine. One strategy’s retained historical regression matched its sealed reference across 228 session decisions and 225 modeled trade lifecycles.

Wilbur Powerhouse | Mechanical Design, year-long team capstone

- Designed and iterated the suspended planetary sculpture’s parametric OpenSCAD assembly, carrier and wheel interfaces. Widened the wheel path while retaining the tooth profile; contributed to CNC preparation and nominal interference checks.

Formsby | Creator, search prototype

- Built a Next.js and PostgreSQL prototype that turns a structured request into a queued search, assesses and ranks listings, and saves results for signed-in users.

EDUCATION & TOOLS

Lehigh University | B.S. Mechanical Engineering, May 2026

Materials Science and Engineering minor expected December 2026. SOLIDWORKS CAD Design Associate (CSWA), Dec. 2024.

Python, JavaScript, React/Next.js, SQL/PostgreSQL, REST APIs, Git, automated testing, OpenSCAD, SOLIDWORKS, AutoCAD, Excel macros, Minitab.